

Sheet - 11 Exception Handling

1. Write a Python program to calculate the result of dividing a person's age by zero and handle the potential division by zero error.

2. Write a Python program that attempts to concatenate a string (name) with an integer (rollNo) using the + operator inside a try-except block. Handle the potential TypeError raised due to the invalid operation.

3. Write a Python program that defines a function check(num) to perform the following operations:

- If the input number num is even, concatenate the string "1" with the number num and print the result.
- If the input number num is odd, attempt to divide the number num by zero and print the result.

Handle the potential TypeError and ZeroDivisionError exceptions raised by the function calls inside a try-except block.

4. Divide by Zero Error Handling:

Write a Python script that takes two numbers as input from the user and divides the first number by the second number. Handle the ZeroDivisionError exception gracefully and print an appropriate message if the second number is zero.

5. Type Error Handling:

Write a Python function that takes two arguments and concatenates them. Handle the TypeError exception if one of the arguments is not a string and print an appropriate message.

6. Index Error Handling:

Write a Python script that attempts to access an element of a list using an index specified by the user. Handle the IndexError exception if the index is out of range and print an

appropriate message.

7.Value Error Handling:

Write a Python script that prompts the user to enter an integer. Handle the ValueError exception if the input cannot be converted to an integer and print an appropriate message.

8.Custom Exception Handling:

Define a custom exception class called CustomError. Write a Python script that raises this exception under a specific condition and handles it using a try-except block.

9.Multiple Exception Handling:

Write a Python script that performs a series of operations (e.g. dividing two numbers) and handles multiple exceptions using separate except blocks for each exception type.

10.Raising Exceptions:

Write a Python script that prompts the user to input a number and performs the following operations:

Converts the input to an integer.

Raises a ValueError with the message "Error: Entered Negative Numbers" if the input number is less than or equal to zero.

Handles the potential ValueError exception using a try-except block and prints the error message if it occurs.

MCQ

1. What is the purpose of exception handling in Python?

- a. A) To prevent errors from occurring in the code
- b. B) To terminate the program when errors occur
- c. C) To gracefully handle errors that occur during program execution
- d. D) To ignore errors and continue program execution

2. Which keyword is used to raise an exception in Python?

- a. A) break
- b. B) continue
- c. C) raise
- d. D) catch

3. In a try-except block, where is the code that may raise an exception placed?
 - a. A) Inside the try block
 - b. B) Inside the except block
 - c. C) Outside both try and except blocks
 - d. D) Inside the finally block
4. Which exception is raised when an invalid mathematical operation is performed, such as dividing by zero?
 - a. A) ValueError
 - b. B) TypeError
 - c. C) ZeroDivisionError
 - d. D) ArithmeticError
5. What is the purpose of the finally block in a try-except statement?
 - a. A) It executes if an exception occurs
 - b. B) It executes regardless of whether an exception occurs or not
 - c. C) It executes only if no exception occurs
 - d. D) It is not necessary in a try-except statement
6. Which of the following is NOT a standard Python exception?
 - a. A) FileNotFoundError
 - b. B) ZeroError
 - c. C) TypeError
 - d. D) IndexError
7. In exception handling, what does the except keyword do?
 - a. A) Catches and handles exceptions
 - b. B) Raises an exception
 - c. C) Terminates the program
 - d. D) Ignores exceptions
8. Which of the following is NOT a best practice in exception handling?
 - a. A) Catching specific exceptions rather than using a generic except block
 - b. B) Raising exceptions for expected error conditions
 - c. C) Using a single try-except block for the entire program
 - d. D) Providing informative error messages
9. What does the else block in a try-except statement do?
 - a. A) It executes if an exception occurs
 - b. B) It executes regardless of whether an exception occurs or not
 - c. C) It executes only if no exception occurs
 - d. D) It is not used in a try-except statement

10. Which statement is used to handle multiple exceptions in a single except block?

- a. A) elif
- b. B) except all
- c. C) except (Exception1, Exception2)
- d. D) except (Exception1 and Exception2)